

Sanatan Kumar Gupta

+91 8866113558 | sanatangupta@outlook.com | linkedin.com/in/sanatangupta | codeforces.com/profile/iwnl

EDUCATION

Indian Institute of Technology, Roorkee

Integrated Master's Degree in Mathematics and Computing

Jul 2024 – Ongoing

CGPA : 8.08 | [Transcript](#)

Kendriya Vidyalaya, Surat

All India Senior School Certificate Examination

Jun 2024

Overall Percentage: 93.2% | [Result](#)

ACHIEVEMENTS

- **Codeforces - Expert**, Max. Rating: **1758** (top 500 in India) [Profile](#)
- **AtCoder - 8Kyu**, Max. Rating: **420** [Profile](#)
- **JEE Advanced 2024**: AIR 4321 (top 2.4%)
- **JEE Main 2024**: AIR 3408 (top 0.21%)
- Qualified **Indian Olympiad Qualifier in Mathematics (IOQM)** 2024
- **INSPIRE SHE Scholarship** Recipient

PROJECTS

Custom Memory Allocator

Oct 2025 – Dec 2025

- Built a heap allocator in C++ supporting malloc/free with free-list management, block splitting and coalescing
- Handled memory alignment and minimized fragmentation across diverse allocation patterns
- Benchmarked against glibc malloc to evaluate throughput and memory efficiency under varying workloads

Concurrent In-Memory Key-Value Store

Dec 2025 – Jan 2026

- Built a thread-safe in-memory key-value store in C++ using RAII for deterministic resource management
- Designed storage using hash maps with read-write locks to support safe concurrent access
- Exposed a clean API supporting CRUD operations with proper encapsulation and error handling

High-Performance Order Matching Engine

Jan 2026 – Present

- Developed an order matching engine in C++ supporting limit and market orders with price-time priority
- Used ordered maps and priority queues to manage bid/ask books efficiently
- Implemented logging and validation to ensure correctness under high-frequency order flows

TECHNICAL SKILLS

- **Languages:** C, C++, Python, R, Julia
- **Systems:** Memory Management, Concurrency, Multithreading, RAII, Cache-Aware Design
- **CS Fundamentals:** Data Structures & Algorithms, OOP, Computer Network
- **Tools & Env:** Git, Linux, GDB, Valgrind, L^AT_EX, NumPy

COURSEWORK

- **Core:** Real Analysis, Stochastic Calculus, Discrete Mathematics, Operations Research, Complex Analysis*
- **CS:** Data Structures & Algorithm, Object-Oriented Programming, Data Science, Database Management Systems*, Theory of Computation*

POSITIONS OF RESPONSIBILITY

- **Editorial Member, Geek Gazette:** Write and curate content for the official technical magazine of IIT Roorkee.
- **Editorial Member, Watch Out! News Agency:** Contribute to campus journalism and reporting for IIT Roorkee's official news body.

**Ongoing courses*